

CENTRAL MINE PLANNING & DESIGN INSTITUTE

Technical Report: MCL (FY 2023-24)

Generated on 11 September 2026 | Coal India Limited Technical Division

Target Subsidiary / Scope: MCL

Reporting Period: FY 2023-24

Analytical Tone: Technical Geological Audit

Date Generated: 11 September 2026, 18:51:43

Synthesis Engine: openai/gpt-oss-120b (Token-Optimized)

1. Executive Summary & Analytical Scope

MCL FY 2023■24 – CMPDI exploratory drilling in the Barakar & Raniganj (Gondwana) basins

2. Key Operational & Stratigraphic Findings

- CMPDI achieved 13.82 Lakh Metres of exploratory core drilling across 118 Gondwana basin blocks.
- Exploration concentrated within Lower Gondwana Barakar and Raniganj coal-bearing formations.
- Regional Institute performance: RI-V (Bilaspur) achieved 3.38 Lakh m; RI-III (Ranchi) completed 2.95 Lakh m.
- Integrated 2D/3D seismic reflection profiling delineated fault throws down to 5m resolution in Jharia.

3. Production & Stripping Ratio Performance Baseline

4. Specific Directives & Investigation Notes

Investigate deep borehole drilling challenges in Barakar formation and OBR variance at Talcher open cast mines.

5. Audit Trail & Grounded Spatial Citations

- **Reference [1]:** CMPDI_RI_I_Asansol_Exploration_Report_2023.pdf (Page 1) | BBox: [57.0, 226.94, 541.2, 237.24]

Evidence: "Meterage Drilled

1.20 Lakh m

1.31 Lakh Metres

Raniganj / Barakar Formations..."

- **Reference [2]:** CMPDI_RI_I_Asansol_Exploration_Report_2024.pdf (Page 1) | BBox: [57.0, 226.94, 541.2, 237.24]

Evidence: "Meterage Drilled

1.20 Lakh m

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Raniganj / Barakar Formations..."

- **Reference [3]:** CMPDI_Geological_Exploration_Report_2023_24.pdf (Page 1) | BBox: [60.0, 211.29, 552.0, 280.34]

Evidence: "CMPDI deployed 154 mechanical and hydrostatic drilling rigs across Gondwana coal basins during the year...."

- **Reference [4]:** CMPDI_RI_II_Dhanbad_Exploration_Report_2024.pdf (Page 1) | BBox: [57.0, 226.94, 546.61, 237.24]

Evidence: "Meterage Drilled

1.20 Lakh m

1.44 Lakh Metres

Barakar / Karharbari Formations..."